5. Apply EM algorithm to cluster a set of data stored in a .CSV file. Use the same data set for clustering using *k*-Means algorithm. Compare the results of these two algorithms and comment on the quality of clustering. You can add Java/Python ML library classes/API in the program.

```
# -----Kmeans----

from sklearn import datasets

from sklearn import metrics

from sklearn.cluster import KMeans

from sklearn.model_selection import train_test_split


iris = datasets.load_iris()

X_train,X_test,y_train,y_test = train_test_split(iris.data,iris.target)

model =KMeans(n_clusters=3)

model.fit(X_train,y_train)

kmean=metrics.accuracy_score(y_test,model.predict(X_test))

print("acuracy of Kmean",kmean)

#-------Expectation and Maximization----------

from sklearn.mixture import GaussianMixture

model2 = GaussianMixture(n_components=5)

model2.fit(X_train,y_train)

model2.score

em=metrics.accuracy_score(y_test,model2.predict(X_test))

print("acuracy of EM",em)
```